

R.O.B. VISION

VIRTUAL ROBOT COMPANION / MK.01



USER GUIDE

Play, preview, setup, and problem solving

VERSION 0.1.7

UNO Q · RETROPIE · BATOCERA

CHAPTER 01 / STORY

Meet Buddy

Buddy lived in the dark between game frames. A flash told him to turn; another told him to lift. He could move the pieces, but the edge of the screen was as far as he had ever gone.



Then a little grid of lights appeared beyond the game. Buddy blinked back. The lights became eyes on the UNO Q, and a larger window opened. For the first time, he could see his hands, the game table, and someone waiting for him to move.

Now each signal gives Buddy another step into the room. When the game ends, he waits for the next round.

CHAPTER 02 / USER GUIDE

R.O.B. Vision user guide

The UNO Q controls Buddy, the virtual robot shown in Mission. RetroPie or Batocera sends Gyromite and Stack-Up commands through the game-frame link. His movements and game pieces appear in your browser.

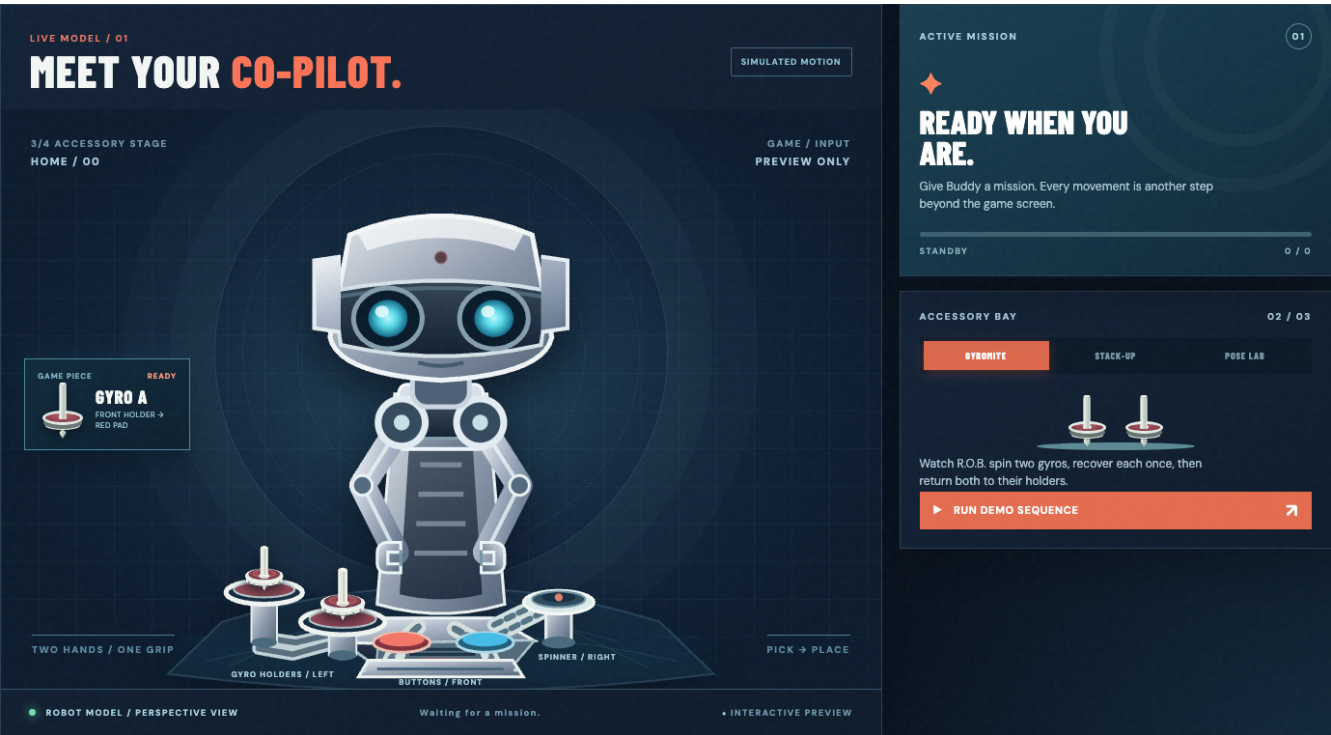
Open R.O.B. Vision

Install R.O.B. Vision with the two commands in the [installation guide](#): one on the UNO Q, one on the console. The console command prints first-time pairing details for Setup and configures its receiver automatically.

- 1. Open your UNO Q's . local address or LAN IP in a browser on the same network.
- 2. If you see the app chooser, choose **R.O.B. Vision**.
- 3. Open **Mission** to see Buddy, his accessories, and the Game Table.

Apps returns to the chooser when both controller apps are installed. Finish a game before changing apps manually. Starting a registered game selects Buddy automatically; you do not need a browser open for game commands to reach him.

Status	What it tells you
GYROMITE / CONNECTED	Mission is connected to the UNO Q with Gyromite selected.
RETROPIE ONLINE or BATOCERA ONLINE	Your console's receiver has checked in recently.
GAME FRAMES LINKED	Signals from the selected game are reaching Buddy.



Try a demonstration

When no supported game is running, select **Gyromite** or **Stack-Up** and **Run Demo Sequence**. Gyromite demonstrates a held unspun gyro, two spinning gyros on pads, spin-down, re-spin, and return to holders. The illustrative spin clock is 55 seconds. Stack-Up moves red alone and then a blue/white group. **Home** resets the pieces. Pose Lab is a character-only movement sandbox. A live game launch stops the demo and selects that game's scene.

Choose an NES emulator

For automatic R.O.B. movement on RetroPie, select **lr-robvision-fceumm** or **lr-robvision-nestopia** for the game. These choices let the game send Buddy its commands. On supported Batocera, the installer selects the R.O.B. Vision FCEUmm wrapper for registered games; Nestopia is available as a per-game choice. See the [RetroPie](#) and [Batocera](#) setup guides.

Play Gyromite

Open [Setup](#) to check the console and game-frame status. Start Gyromite and watch Mission for **GAME FRAMES LINKED**. R.O.B. follows complete commands from the game. A spinning virtual gyro on a pad keeps its gate pressed while R.O.B. moves elsewhere; an unspun gyro can press one gate while held there. The matching gate on the game screen should move.

Fast Gates responds immediately while Gyromite is selected. Tap **Lower Blue** or **Lower Red** to press, tap again to release, or choose **Release Both**. On Mission, **2** toggles blue, **1** toggles red, and **0** releases both while you are not typing in a field. Holds expire after 60 seconds; both buttons release on game exit or lost receiver data. Buddy's hands must be empty, the matching gyro must be on its holder, and the target pad must be free. Return the pieces with regular controls or choose **Home** to reset the gyros. Watch the game screen to confirm the gate's response.

Play Stack-Up

The live model begins with five colored blocks on Tray 3. In Direct mode, jumping Hector onto a command tile sends **Left**, **Right**, **Up**, **Down**, **Open**, or **Close** through the frame link. Manual controls use the same virtual model. A lower grip carries the contacted block and all blocks above it. Impossible moves are rejected and leave pieces in place. **Home** restores the initial stack.

Current game limits

Stack-Up's virtual arrangement does not score a round or initialize Bingo's distinct starting layout. Use the game screen to check its target and result. Stack-Up does not use Gyromite's gate-button return path.

Test mode and reset

Launch the game through its R.O.B. Vision emulator choice and enter Test mode. The game-frame link makes R.O.B.'s red light blink automatically on Mission and Setup when it recognizes the Test signal. A separate ready-light command makes the light steady briefly. Exiting Test mode returns the UNO Q matrix to the game's eyes. Neither signal moves R.O.B. **Preview Red Light** on Setup demonstrates the animation without claiming a game signal.

Home resets the selected virtual game. Live **Emergency Stop** clears the game selection; in the offline preview it cancels the script until reset. A browser reload restores the UNO Q's current snapshot. See [Setup](#), [gameplay](#), and [troubleshooting](#).

CHAPTER 03 / USER GUIDE

R.O.B. Vision gameplay guide

The UNO Q updates Buddy and the game pieces shown in your browser. A paired RetroPie or supported Batocera console sends movement commands from the game; Gyromite's virtual pad states return to that console's Controller 2.

Gyromite

The [original Gyromite booklet](#) describes **Test**, **Direct**, **Game A**, and **Game B**. Direct sends six R.O.B. actions: up, down, left, right, open, and close. In Game A, Start switches into the robot-transmission screen for a command and then back to the professor. In Game B, the professor walks while Controller 1 directly commands R.O.B. A gyro on the blue or red tray lowers the matching gate.

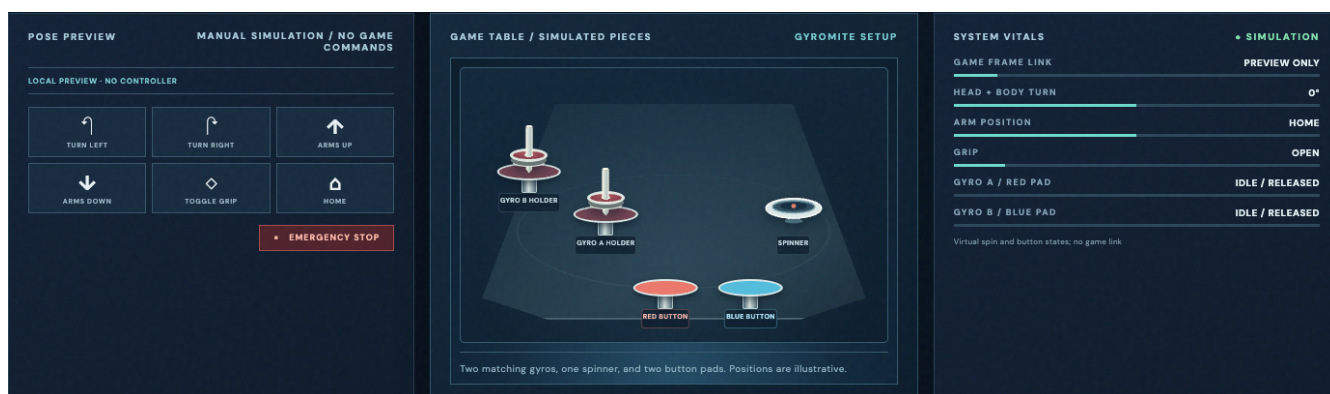
The manual explicitly says a gyro **does not need to spin to operate only one gate**. R.O.B. can hold an unspun gyro on the virtual pad, then lift it to release that pad. If both gates must be controlled, spinning one gyro lets it stay upright on one pad while R.O.B. uses his hands to put the other on the other pad. A gyro that is no longer needed belongs on its holder. The spinner is a means to free R.O.B.'s hands, not an obligatory step for every pad press.

Buddy starts at **Home**, level 6. Send **Down** twice to reach level 2, where his hands meet either holder, the spinner, and the colored pads. **Close** picks up the gyro. Raise it once to level 4 before moving left or right, then lower it to level 2 at its destination.

The table is shown in perspective, so accessories may look higher or lower even though they use the same working level.

Try the demo: with no game running, select Gyromite and **Run Demo Sequence**. Buddy presses a pad with an unspun gyro, uses two spinning gyros, and returns them to their holders. The sequence stops at **COMPLETE**. Select **Home** to reset. The demo's spin lifetime is illustrative.

During play: the UNO Q tracks both gyros and sends red and blue pad states to the active console. It releases both buttons on game exit or a lost link. **Fast Gates** presses or releases a matching pad immediately; each hold expires after 60 seconds. Watch the game screen for Hector and the gate response.



Pose Preview, Game Table, and System Vitals. Fast Gates appears during live Gyromite play.

Stack-Up

The [Stack-Up booklet](#) describes Direct, Memory, and Bingo. Buddy's Game Table has five trays and five colored blocks. Direct and Memory start with all five on Tray 3, top to bottom **red, white, blue, yellow, green**. In Direct mode, move Professor Hector onto the **left, right, up, down, open, or close** command keys. Memory plays a programmed command series; Bingo sends a command when one row or column completes. Buddy can carry a block with all the blocks above it.

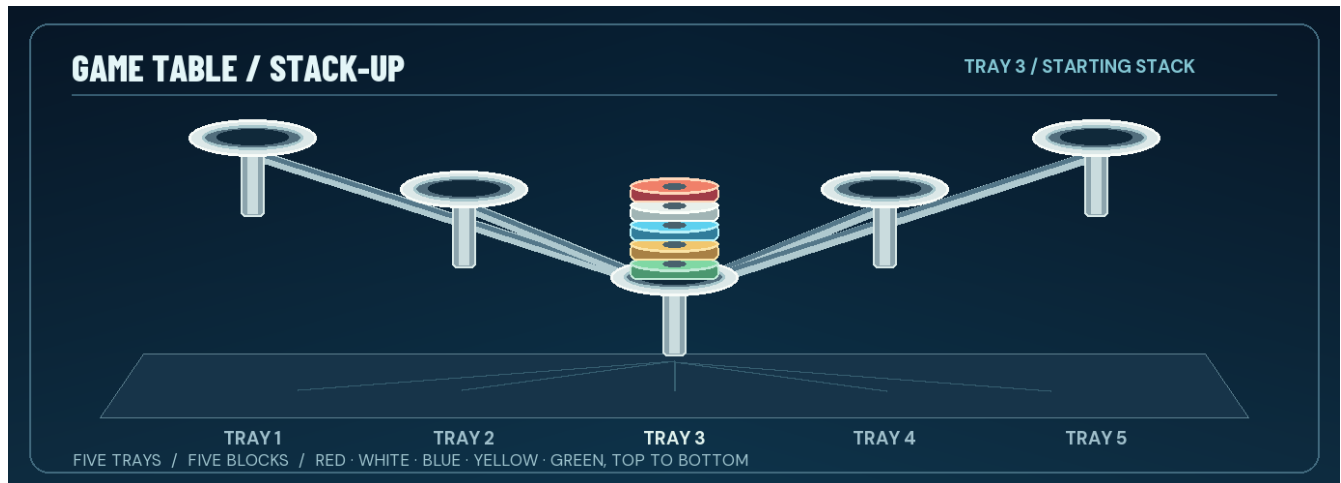


Illustration of the initial Stack-Up Game Table, using the dashboard's tray layout and block order.

Try the demo: with no game running, select Stack-Up and **Run Demo Sequence**. Buddy moves red alone, then carries blue and white together. All five blocks remain visible in the Live Model, Accessory Bay, and Game Table. The **Stack-Up Commands** buttons apply one virtual action at a time. **Home** restores the starting stack. If a move is blocked, the pieces stay in place and the Activity Feed explains why.

Current limits: Buddy starts with the Direct/Memory stack in every Stack-Up mode; Bingo's different starting arrangements are not modeled. Compare the virtual arrangement with the game's target on its screen. The game sends movement commands but does not tell Buddy its target, score, or whether the round was completed. Stack-Up does not use Gyromite's gate-button return path.

Pose Lab and recovery

Pose Lab lets you try Buddy's head, body turn, arms, and shared grip. Its short demo is a greeting. If the browser reconnects during a game, it resumes from the UNO Q's current state. If the Gyromite link drops, both virtual gate buttons release.

CHAPTER 04 / USER GUIDE

Get Buddy ready to play

Use R.O.B. Vision **Setup** to pair a console, register your game filenames, and check Buddy's controls. Open R.O.B. Vision at your UNO Q address, then choose **Setup**.

Installing for the first time? Start with [Installation and Setup](#).

Pair a console

1. Keep the console installer open when it shows a six-digit code and certificate fingerprint.
2. In Setup **Console Link**, enter the console's `.local` name or LAN IP address.
3. Enter the code and fingerprint printed by the installer.
4. Choose **Pair Console**.
5. Wait for the console terminal to say **Pairing complete**.
6. Choose **Check Link** in Setup.

The pairing window lasts five minutes. If it expires, open a new window with the console installation command. Choose **Pair Another Console** to add another machine.

Each paired console appears in its own row. **Receiver Waiting** means this UNO Q has not heard from its receiver recently. Check the console and connection; the message alone does not mean the machine is switched off.

A console uses one UNO Q at a time. Pairing it to another UNO Q moves its connection there. Upgrades keep saved pairings.

Register your game filenames

If your game has a different filename from the supplied entries, add it to that console's registry.

1. Exit the game.
2. In **Console Link**, choose **Edit ROMs** beside the console.
3. In the **Game Registry** editor, find the games section.
4. Add the game's exact filename, including its extension, with `gyromite` or `stack_up` as its value.
5. Choose **Validate**, then **Save to Console**.
6. Launch the game again.

For example, an entry for your own Gyromite ZIP can look like this:

```
"My Gyromite.zip": "gyromite"
```

Add the entry inside the existing `games` object and keep the other entries. Each console has its own registry. Changes and upgrades keep your saved filenames.

Check the game connection

1. On RetroPie, launch the game with `lr-robvision-fceumm` or `lr-robvision-nestopia`. On Batocera, launch the registered game with its installed R.O.B. Vision emulator integration.

2. In Setup, check that the correct game is shown.
3. Look for **GAME FRAMES LINKED**.
4. Send one movement command from the game and watch Buddy on Mission.

GAME FRAMES LINKED means game signals are reaching R.O.B. Vision. A blocked movement is explained in Mission's **Activity Feed**.

Try the game's Test mode

1. Enter **Test** in the game. In Stack-Up, press Select on the ROBOT BLOCK screen, choose Test, then press Start.
2. Watch Buddy's red light on Mission or Setup. It should blink automatically.
3. Check the UNO Q display for a pulsing T.

Test checks the connection without moving Buddy. A brief steady red light acknowledges the game's ready signal. **Preview Red Light** lets you try the animation yourself; it does not check the game connection.

Check Gyromite's gates

1. Enter Gyromite **Game A** and leave a colored gate visible.
2. In Setup, choose **Lower Blue** or **Lower Red** for that gate.
3. Watch the game screen for the gate to move.
4. Choose **Release Both** when finished.

Choose a color again to release it. A hold also releases after 60 seconds. If a gyro is already in use, return it with the normal controls or choose **Home** to reset the pieces before using Fast Gates.

Check Stack-Up movement

In Setup, try **Left**, **Right**, **Up**, **Down**, **Open**, or **Close** one at a time. Watch Mission's Live Model and Game Table. Raise Buddy's hands before moving sideways past a stack. A blocked move keeps the pieces intact; read the Activity Feed, raise the hands, and try again.

Remove a console

1. In the console's row, choose **Remove**.
2. Choose **Confirm Remove**.

Other pairings stay saved. To pair that console again, rerun its console installation command with `--pair` added at the end.

For physical gamepads and Buddy's Player 2 controls, see the [Controller Router Guide](#). For game rules and pieces, see the [Game Manual](#).

CHAPTER 05 / USER GUIDE

Controller Router: get your controllers ready to play

Controller Router lets you choose which gamepad controls each player, and which systems use those choices. It also gives VirtualGlove and Buddy a shared home on your UNO Q.

Already happy with your controls? Keep your saved choices. You only need this guide when you want to change a player, choose a system, or troubleshoot a pad.

Open Setup

1. In your browser, open your UNO Q's address, such as `http://your-uno-q.local/`.
2. Choose **Setup** on the Controller Router page.
3. Select your paired console under **Console connection**.

You will see **Players** first, then **Systems**. Each console keeps its own settings. If no console appears, pair it in your controller app's Setup page first.

Choose who plays

Before assigning a gamepad, configure its buttons in EmulationStation, your console's game menu.

1. Exit any running game.
2. In Router **Setup > Players**, find your gamepad by name.
3. Choose **Player 1**, **Player 2**, **Player 3**, or **Player 4**. Choose **Unassigned** if you do not want that pad to join a Router player.
4. If VirtualGlove is installed, choose its gesture player under **VirtualGlove player**, or leave it **Unassigned**.
5. Choose **Save assignments**.
6. Launch a game to try the new choices.

Keep a physical gamepad assigned to **Player 1** for the usual menu and exit controls. Several pads can share one player; a pad can belong to only one player. Buddy stays on **Player 2** for Gyromite's red and blue gates.

A game must support additional players for Player 3 or Player 4 to do anything. Assigning four pads does not turn a two-player game into a four-player game.

Choose which systems use Router

Want Router for NES but your normal controls for Mega Drive / Genesis? You can choose each system separately.

1. With the game closed, open **Setup > Systems**.
2. Under **Routing selection**, choose **Choose systems individually**.
3. For each listed system, choose one of the following options.
4. Choose **Save assignments**, then launch your next game.

Choose	What happens in a game
Controller Router	The system uses the player assignments you saved in Players , with the buttons you configured in EmulationStation.
My existing setup	The system uses its normal controller setup. Router adds no player-routing changes for that game.

Buddy's games and VirtualGlove require Controller Router enabled for NES. You can still choose **My existing setup** for NES when you want to use your own controls instead.

New Router setups start with **NES only**. Upgrades keep your choices. **All Libretro systems, including newly added systems** applies Router to every supported RetroArch system. In individual mode, a newly discovered system starts with **My existing setup**.

The list covers systems that offer supported RetroArch emulators. Separate emulators, such as a standalone Amiga emulator, keep their own controls. Your choice also stays saved if a system is temporarily unavailable.

Check a gamepad

1. In Router Setup, choose **Test inputs**.
2. Press a direction or button on the pad during the next five seconds.
3. Read the result, then repeat for another pad if needed.

No button presses detected can simply mean you did not press anything during the test. Try again before changing settings.

Status	What to do
Connected	The pad is available. Try Test inputs to identify it.
Unavailable	Turn on or reconnect the saved pad. If you replaced it, configure and assign the new one.
Mapping refreshed or updated	Router found your newer EmulationStation button setup. Exit and relaunch the game to use it.

Reload brings back the saved choices and discards unsaved edits. **Restore previous** restores the preceding saved setup. Exit the game before saving or restoring.

Start a game or choose an app

Launch a registered game on your paired console. Router selects VirtualGlove or R.O.B. Vision automatically, so you do not need a browser open for the controller to join the game. Only that app supplies game input and display cues. When the game ends, the UNO Q returns to its neutral display.

To open an app yourself, visit the UNO Q address. With one app installed, it opens directly. With both installed, choose VirtualGlove or R.O.B. Vision. **Apps** in either app returns to the chooser; it appears only when both are installed. Finish a game before changing apps manually. Both apps stay available in the background.

After a reboot, the UNO Q waits for a game or your choice. You do not need to reinstall or pair again.

Get back to your game

My pad works in the game menu but not in the game. Try another connected pad: yours may be assigned to Player 2. Exit the game, check **Players**, and use **Test inputs** to identify the pad you want on Player 1. Save and relaunch.

A wireless pad went to sleep. Wake it or reconnect it. Router keeps its player controllers connected and returns the pad to its saved player. If the Router service itself restarted, exit and relaunch the game once it is ready.

My buttons changed after I remapped them. Exit the game, confirm the new buttons in EmulationStation, and relaunch. Your player assignment stays saved.

I cannot save. Finish the running game first. If another page changed the settings, choose **Reload**, make your choices again, and save.

The UNO Q is showing the wrong app. Check that the game is registered in the right app and that the console is paired with this UNO Q. Exit the game before making changes.

Check Buddy's gates

1. Launch Gyromite in **Game A** and stop where a colored gate is visible.
2. In R.O.B. Vision **Setup**, choose **Lower Blue** or **Lower Red** for that gate.
3. Watch the game screen, then choose **Release Both**.

Your physical Player 1 pad controls Hector. Buddy's Player 2 input controls the gates. Stack-Up uses game commands to move Buddy and the blocks; it has no gate buttons.

The **Controllers** button beside a paired console in R.O.B. Vision Setup opens that console's assignment editor. For pieces and movement, see the [Game Manual](#). For connection problems, see [Troubleshooting](#).

CHAPTER 06 / USER GUIDE

Install and set up R.O.B. Vision

Install R.O.B. Vision on your UNO Q and console, pair them, then try your first game. **Installation and upgrades use the same command on each device.** Upgrades keep your saved pairing and game filenames.

Before you begin

You need an Arduino UNO Q with App Lab, a RetroPie or Batocera console, a physical gamepad configured in EmulationStation, and your own Gyromite or Stack-Up game files. Put both devices on the same trusted network and connect them to the internet for installation.

Exit any running game. On RetroPie, also exit EmulationStation to its terminal before installing: replacing virtual controllers while the game menu is open can make it crash.

The installer checks the required tools and emulator support. If it reports a missing requirement, follow that message and rerun the same command. Detailed platform requirements are in the [Configuration Reference](#).

1. Install or upgrade the UNO Q

In the UNO Q terminal, signed in as `arduino`, install the app:

```
curl -fsSL https://github.com/mathan416/ROB-Vision/releases/latest/download/install.sh | sh -s -- -u no-q
```

Leave the terminal open until installation finishes. The first shared Matrix build may take several minutes.

The installer prints links using both the UNO Q's `.local` name and LAN IP address. Open its **Setup** link on your computer, tablet, or phone.

Controller Router is included. Visiting the UNO Q address opens R.O.B. Vision when it is the only controller app installed, or shows the app chooser when VirtualGlove is installed too. **Apps** returns to the chooser. Both apps stay available; finish a game before changing apps manually.

2. Install or upgrade the console

In the console terminal, signed in as `pi` on RetroPie or `root` on Batocera, install the console software:

```
curl -fsSL https://github.com/mathan416/ROB-Vision/releases/latest/download/install.sh | \
sh -s -- -c console your-uno-q.local
```

Replace your `-u no-q .local` with the UNO Q name or IP address printed by its installer. The same command detects RetroPie or Batocera and includes Controller Router and the game integration.

For a new pairing, leave the terminal open. It prints a **six-digit code** and a **certificate fingerprint**, then waits up to five minutes for you to pair. An upgrade keeps the existing pairing.

3. Pair the console

1. In R.O.B. Vision Setup, find **Console Link**.
2. Enter the console's name or IP address.
3. Enter the code and fingerprint from the console terminal.

4. Choose **Pair Console**.
5. Wait for the console to say **Pairing complete**.
6. Choose **Check Link**.

Buddy is assigned to Player 2. Existing physical-controller assignments stay saved. You do not need another terminal command after pairing.

4. Check players and game filenames

1. Open **Setup** from the Controller Router page at the UNO Q address.
2. Select your console and check **Players**. Keep your physical gamepad on Player 1; Buddy stays on Player 2.
3. Under **Systems**, keep **Controller Router** enabled for NES. Other systems can use **My existing setup** if you prefer your normal controls.
4. With the game closed, choose **Save assignments** if you changed anything.
5. In R.O.B. Vision Setup, choose **Edit ROMs** beside the console if your filename differs from the supplied entries.

The [Setup Guide](#) explains how to add a filename. The [Controller Router Guide](#) explains pad tests and system choices.

5. Try your first game

1. Launch Gyromite or Stack-Up. On RetroPie, use **lr-robvision-fceumm** or **lr-robvision-nestopia**. Batocera uses the installed R.O.B. Vision integration for registered games.
2. Open Mission and look for the correct game and **GAME FRAMES LINKED**.
3. Enter the game's **Test** mode. Buddy's red light should blink and the UNO Q should show **T**.
4. Enter **Direct** mode and send a movement command. Watch Buddy respond.
5. For Gyromite, enter **Game A** and try one colored gate with Setup's **Lower Blue** or **Lower Red**, then **Release Both**.

Starting a registered game selects Buddy's controller automatically. A browser is optional for sending game commands, but keep Mission open to see Buddy's pieces and position.

Upgrade, repair, or pair again

To upgrade or repair, close games and repeat the same installation command on each device. Keep the printed backup paths until you have checked a game. Upgrades keep your pairings, filenames, and player assignments.

If you removed a console in Setup, add `--pair` to the end of its console command to open a new pairing window. Use **Pair Another Console** to add a different machine. Each console connects to one UNO Q at a time.

See the [Game Manual](#) for playing, or [Troubleshooting](#) if a check does not work.

CHAPTER 07 / USER GUIDE

Get back to playing with Buddy

Start with the symptom you can see. Try one check, then test the game again.

The website will not open

1. Open your UNO Q's . local address on the same network.
2. If the name does not work, try the LAN IP link printed by its installer.
3. Choose R.O.B. Vision if the Apps page offers both controllers.

If neither address works, check the UNO Q's power and network connection. If an installation failed, keep its printed error and rerun the same installation command after fixing the reported problem.

The console says Receiver Waiting

1. Turn on the console and check that it is connected to the same network.
2. In R.O.B. Vision Setup, choose **Check Link** beside that console.
3. Confirm the console is paired with this UNO Q.

A console connects to one UNO Q at a time. If you paired it with another, use that UNO Q or pair it here again.

Receiver Waiting means there has been no recent receiver check-in; it does not prove the console is off.

The game is running but Buddy does not move

1. Check that Setup shows the right game.
2. Look for **GAME FRAMES LINKED**.
3. On RetroPie, confirm the game uses **lr-robvision-fceumm** or **lr-robvision-nestopia**. On Batocera, use its installed R.O.B. Vision integration.
4. Send one command in the game's **Direct** mode.
5. Read Mission's **Activity Feed**.

A blocked command can mean Buddy needs to raise his hands or move to the correct holder or tray. Follow the message and try again. If no game is selected, check its filename with **Edit ROMs** in Setup, then relaunch.

Gyromite's gate does not move

1. Enter **Game A** and leave the matching colored gate visible.
2. In Setup, choose **Lower Blue** or **Lower Red**.
3. Watch the game screen, then choose **Release Both**.
4. If it stays still, exit the game and open Router **Setup > Players**. Check your physical pad is Player 1 and Buddy is Player 2.
5. Under **Systems**, check NES uses **Controller Router**, then relaunch.

If Fast Gates reports that a piece is busy, return it with the normal controls or choose **Home** to reset the gyros. Fast Gates appears on Mission only during live Gyromite play.

A gate stays down

Choose **Release Both**. Fast Gates also releases after 60 seconds. A gyro resting on a pad can keep the gate pressed during normal play; lift that gyro to release it.

A Stack-Up move is blocked

Read the **Activity Feed**. Raise Buddy's hands above the destination stack before moving sideways. Open hands can lower around the current stack to select a block; closed hands cannot lower into it.

A blocked command leaves the blocks intact. Raise the hands and try again. Choose **Home** only when you want to restore the starting stack.

My gamepad works in the menu but not in the game

Try another connected pad first: the one you are holding may be Player 2. Exit the game and check the named pads in Router **Setup > Players**. Choose **Test inputs**, press a button, then correct the assignment if needed. Save and relaunch.

If one game still picks the wrong player, check whether you saved different controller choices in that game's RetroArch settings. Those can take priority over Router. Keep unrelated game settings. The [Technical Reference](#) explains how these settings interact.

A wireless pad fell asleep or woke during play

Wake or reconnect it. Router keeps its player controllers connected and returns your pad to its saved player. If the Router service itself restarted, exit the game and relaunch once it is ready.

Test mode does not blink the red light

Check the correct game and **GAME FRAMES LINKED**, then enter the game's **Test** mode. **Preview Red Light** only demonstrates the animation; it does not test the game connection.

The demo will not start

Exit the running game first. Demos work while the console is paired and idle. Choose Gyromite or Stack-Up on Mission, then **Run Demo Sequence**. **Home** restores the starting pieces.

RetroPie's game menu crashed during installation

Return to the console terminal and rerun the installer with EmulationStation closed. Keep it closed until installation finishes, then reopen it.

Ask for help

Include the game, console platform, emulator choice, visible status, and the Activity Feed message. Describe what you tried and what happened. Keep pairing codes, tokens, private backups, and ROM files out of your report.